Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 39: Getting From Here to There

Walks, Paths, and Connectivity

Chartrand & Zhang, Section 12.2, pp. 444–455

Name: _____ Date: _____

The last lecture of new material

"The graph is in one piece" is a picture, not a definition. Today we make it precise, and prove the small theorem that makes it checkable.

By the end of today you should be able to

1. Distinguish a walk, a trail, a path, and a cycle.
2. State the definition of a connected graph and count components.
3. Follow the proof that a walk contains a path.

1. Four words

Definitions

A **walk** is a sequence of vertices with consecutive ones adjacent. Repetition is allowed.
A **trail** is a walk with no repeated **edge**.
A **path** is a walk with no repeated **vertex**.
A **cycle** is a closed walk of length at least 3 whose only repeat is the start and end.

Every path is a trail, and every trail is a walk — but not conversely.

Classify each, for the graph with edges $v_1v_2,\ v_2v_3,\ v_3v_4,\ v_4v_1,\ v_2v_4$.

Sequence	What it is
(v_1, v_2, v_3, v_4)	A path (no vertex repeats).
(v_1, v_2, v_4, v_1)	A cycle of length 3.
$(v_1, v_2, v_3, v_4, v_2)$	A trail, not a path — v_2 repeats.
(v_1, v_2, v_1, v_2)	A walk only — the edge v_1v_2 repeats.

2. Connected

Definition 12.24

A graph G is **connected** if for every two vertices u and v, G contains a u–v path. Otherwise G is **disconnected**.
A **component** of G is a maximal connected piece. The number of components is written $k(G)$.

G is connected exactly when $k(G) = \underline{1}$.

Count the components.

Graph	$k(G)$
$V = \{1, 2, 3, 4\}$, $E = \{12, 23, 34\}$	1 — connected
$V = \{1, \ldots, 5\}$, $E = \{12, 34\}$	3 — $\{1, 2\}$, $\{3, 4\}$, $\{5\}$
$V = \{1, \ldots, 4\}$, $E = \emptyset$	4

3. A walk is enough

Theorem 12.26

If G contains a u–v walk, then G contains a u–v path.

Proof. Assume $u \neq v$. Among all u–v walks, let W be one of smallest length. We claim W is a path.

Suppose not. Then some vertex x appears twice in W. Delete the portion of W between the two occurrences of x, along with one copy of x. The result is still a u–v walk, and it is shorter than W — contradicting the choice of W. ■

Two things to notice

This is **proof by contradiction** using a **minimal counterexample** — exactly the well-ordering style from Lecture 30. Among all walks there is a shortest one, because lengths are positive integers. And the payoff is practical: to prove a graph connected, you need only exhibit *walks* between pairs. Walks are easy to find; paths require care. The theorem says you may be sloppy.

4. How many edges does connectivity need?

Theorem

A graph of order n and size m has at least $n - m$ components.

Consequence. If G is connected then $k(G) = 1$, so $1 \geq n - m$, giving $m \geq n - 1$.

A connected graph of order n needs at least $n - 1$ edges. With $n = 6$ that is 5.

The bound is necessary, not sufficient

Having $n - 1$ edges does not make a graph connected. On $\{1, \ldots, 5\}$ the edge set $\{12, 13, 23, 45\}$ has $4 = n - 1$ edges and $k(G) = 2$ — a triangle plus a separate edge.

Enough edges is not the same as the right edges. A connected graph with exactly $n - 1$ edges is a *tree*, the subject of the final week.

5. Where this goes

What graphs are for

Connectivity asks whether a network can fail into pieces. **Distance** — the length of a shortest path — is what routing algorithms compute. **Components** are what a connected-component labeller finds in an image.

Chapter 12 continues with Eulerian and Hamiltonian graphs, which we are not covering; Chapter 13 covers trees. Both build directly on today.

Exit ticket

1. Give a sequence that is a trail but not a path, in any graph you like.

2. A graph has order 8 and size 5. Can it be connected? Explain.

Before the break

Read the Chapter 12 Highlights, pp. 482–484.

Self-check (not collected). Section 12.2, Exercises 1, 3, and 5 — odd-numbered, answers in the back.

Coming up

Homework 13 is due tonight, Friday, November 20, 11:59 p.m.
No class Thanksgiving week.
Homework 14 covers Sections 12.1 and 12.2 and is due Friday, December 4 — the last assignment of the term.
Week of November 30: trees and cumulative review.

Questions I still have